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Design and Experiments of a Three-Fingered Dexterous Hand Based on Biomechanical Characteristics of Human Hand Synergies

Wenrui Chen, Yaonan Wang, Zhilan Xiao, Zilong Zhao, Cuo Yan, and Zhiyong Li

***Abstract*—Hand synergy from neuroscience provides an effective tool for anthropomorphic hands to realize versatile grasping with simple planning and control. This article aims to extend the synergy-inspired design from anthropomorphic hands to general multifingered robot hands by reconstructing mechanically the crucial biomechanical characteristics of hand synergy. The synergy-inspired hands are not necessarily humanoid in morphology but perform primary functions like the human hand. At first, the biomechanics of hand synergy is investigated by combining human hand synergies with the anatomy of the musculoskeletal system. Three biomechanical characteristics of the human hand synergy are explored as a basis for the mechanical simplification of robot hands. Second, based on the three synergy characteristics, a design method of robotic hands is proposed. A three-fingered hand is designed, and some typical grasping and in-hand manipulation functions are analyzed. Finally, a prototype is developed, and experiments of grasping and in-hand manipulation are carried out to validate the effectiveness of the design method.**

***Index Terms*—Biomechanisms, mechanical design, multifingered hands, synergy, underactuation.**

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I. Introduction

ROBOTIC hands have been developed with the aim of matching the human hand in terms of dexterity and adaptation capabilities to equip either a dexterous manipulator as an end-effector or the human being as a prosthetic device. To adapt hands to many kinds of grasping and manipulation tasks, multiple degrees of freedom (DOFs) distributed among several fingers are needed. A classic design approach attempts to closely replicate the appearance and dexterity of human hands with sophisticated designs integrating many actuators and sensors [1], [2]. Since it is overly complex to seek perfect structural and functional anthropomorphism, this approach has resulted in a limited number of real-world applications. Some recent innovations in hand design aim at achieving robust, easily programmable, and economically viable robotic hands while reducing the need for functionality [3] via deceasing reduce the number of actuators, using flexible joints [4] and soft structures [5], [6]. For examples, many popular robot platforms for the end user are built with single-actuator grippers such as the Willow Garage PR2, or multifingered hands with a few actuators for power grasping configurations such as Barrett Hand [7], Robotiq Adaptive Gripper [8], and a three-finger compliant gripper [9]. These hands perform certain grasps robustly with simpler mechanics control, but lose part of the dexterity especially in-hand manipulation ability. There is still a huge gap between dexterity and practicality that needs to be overcome.

One promising approach to reduce this complexity of hands without compromising dexterity is through *synergy*, which is from neuroscience and shows that a continuous subspace of configuration space can be used to approximate everyday human hand tasks [10], [11]. Human hand synergy provides a natural modeling paradigm for robotics [12]. For example, synergies can be used in the planning or control algorithms of robot hands for grasping [13] and manipulation [14], [15]. In particular, this concept can be embedded in the mechanical design of hands via coupling joints, transforming some of control intelligence to physical mechanism [16]. Different types of mechanisms were proposed to enable hardware synergies for anthropomorphic hands [17], [18]. To deal with the force issue in postural synergies, the concept of "soft synergies" was presented and a tool was provided to account for force equilibrium [19]. Based on this theory, the Pisa/IIT hand driven by one motor was developed [20]. To improve the in-hand manipulation ability, an actuator

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was added to the Pisa/IIT hand [21]. Via self-defined synergy matrix, a three-fingered underactuated hand actuated by four motors was designed for grasp stability and dexterity [22]. In our previous article [23], a mechanical implementation method of the complete statistical information of human hand synergies was presented, where the few-order synergies are achieved via actuators, while the high-order ones are implemented with mechanical compliance. The method was used in the design of an anthropomorphic hand [24]. The hand synergy, from the statistical analysis of hand movement data [10], provides an inspiration of the transmission design for underactuated hands to overcome the gap between dexterity and complexity. However, with the different grasping data, there are significant differences in the subspace representing synergy [25]. Although the complexity reduction brought by the synergistic organization of the hand led to encouraging results in grasping, how to exploit the internal mechanism of hand synergies to perform more complex manipulation tasks is still an open issue.

Besides the statistical characteristics, human hand synergies can be explained with biomechanics. In daily life, it is easy to observe the coordinated motion of human hand joints. For example, it is difficult to move the proximal and distal interphalangeal (DIP) joints of the index finger independently. Briefly, the tendons of extrinsic finger flexors and extensors cross several joints. Therefore, tension exerted on a tendon would generate torques at several joints. The corresponding relationship exists between the tendinous connective characteristics and the coordinated characteristics of the human hand during daily grasping activities [26]. The biomechanical factors constrain the extent to which individual joints or fingers can be controlled independently of others [27]. The structure of the musculoskeletal system of the human hand have drawn attention from the robotics field [28]. The anatomically corrected testbed hand was the first try toward replicating human hand on anatomical level [29]. Analyzing the human anatomy to extract the function it provides, an anthropomorphic, antagonistically driven hand was built [2]. To mimic closely the biomechanics of the human hand with artificial joints and ligaments, a highly biomimetic anthropomorphic robot hand was developed [30]. These hands were designed to reproduce the musculoskeletal system structure of the human hand while not reflecting the functional realization of human hand synergy. There are few studies on establishing the internal relationship between statistical characteristics and biomechanical characteristics of human synergy and then guiding the design of robot hands.

This article attempts to excavate biomechanical mechanism of human hand synergy through the analysis of the correlation relationship between the anatomical structure and coordinated characteristics, and then present a design method of robot hands based on biomechanical characteristics of human synergy. Based on the biomechanical mechanism of human synergy, robot hands are not necessarily humanoid in morphology but perform primary functions of human hand. Specifically, a synergy-inspired three-fingered hand as shown in Fig. 1 is designed and implemented based on biomechanical characteristics of human hand synergy for the first time. A preliminary version of the article was presented in [31]. With respect to that article, the present

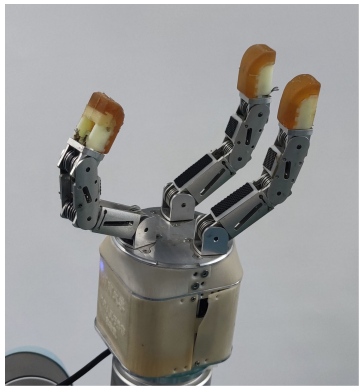


Fig. 1. Synergy-inspired three-fingered hand prototype.

one adds details to the design of the three-fingered hand and the analysis of grasping and in-hand manipulation. Furthermore, more experiments are carried out, including force tests, grasping diversity, and in-hand manipulation ability, to support the effectiveness of the design method. To our knowledge, the main contributions of this article can be summarized as follows:

1) The biomechanical characteristics of hand synergy are extracted first as a basis for the mechanical design of robot hands. Because it does not rely on morphology, this design method has strong transfer ability to general multifingered hands. A three-fingered hand is designed. Its transmission replicates the biomechanical characteristics of synergy even though the morphology is nonanthropomorphic.
2) Design and experiments of the three-fingered dexterous hand proves that it is possible for a synergy-inspired hand to maintain great dexterous grasping and in-hand manipulation ability. Compared with existing typical three-fingered hands, our hand has moderate in weight, size, and the number of motors, with the largest number of joints, and maintain great dexterity and adaptability.

The rest of this article is organized as follows. In Section II, the biomechanical characteristics of hand synergy is explored as a basis for the mechanical design of robot hands. According to the characteristics of hand synergy, a three-fingered hand is designed in Section III, and its kinematic model is developed for analysis of the typical grasping and in-hand manipulation functions in Section IV. Section V carries out experiments to validate the effectiveness of the design method. Finally, Section VI concludes this article.

II. Biomechanical Characteristics of Hand Synergy

The human hand is the best reference for robot hand design. Santello *et al.* [10] investigated human grasping posture data from a large set of daily objects via principal components analysis and revealed that more than 80% of hand posture information is contained in the first two principal components. Error analysis shows that the reconstruction of human high-dimensional data with two principal elements has a small overall error, but the error of a single joint is very large. Especially, the errors of the thumb joints reach up to 80% [32]. Such large errors are unacceptable for grasping and in-hand manipulation. Therefore,

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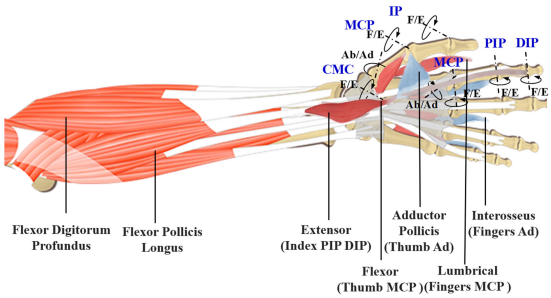


Fig. 2. Neuromuscular control system and joint axes of human hand.

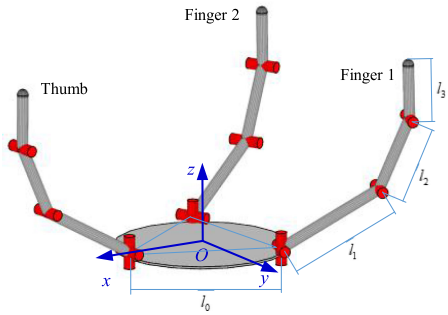


Fig. 3. Joint arrangement of the robot hand.

this article attempts to derive more accurate synergies from the neuromuscular control system of human hand.

In human hand as shown in Fig. 2, the thumb has three joints, namely the interphalangeal, the metacarpophalangeal (MCP), and the carpometacarpal (CMC) joint. The CMC joint is a compound joint. The CMC adduction/abduction (Ad/Ab) axis passes through the proximal end of the metacarpal bone, while the CMC flexion/extension (F/E) axis intersects with the trapezium carpal bone. Each of the fingers has three joints called DIP, proximal interphalangeal (PIP), and MCP joints, respectively. The DIP and PIP joints can generate rotation movement around the axis-F/E while the MCP joint a compound joint whose two orthogonal axes, namely the axis-F/E and the axis-Ad/Ab, respectively.

The statistics of natural hand movements indicates that there is not only correlation but also independence among hand joints [33]. It represents two forms of synergy, namely *coupling* and *grouping*. Its functional link with human musculoskeletal architecture provides a biomechanical basis for hand synergy [34]. Biomechanical characteristics of hand synergy are mainly manifested in the following.

Ad/Ab movement is independent with F/E movement: Since the amplitude of Ad/Ab movement is very small compared to F/E, it is often ignored in the collaborative analysis. However, the remarkable grasping skills of the human hand rely more on the independent Ad/Ab of the fingers than on their independent F/E [35]. Statistically, the Ad/Ab movement is independent significantly with F/E movement [36]. From the anatomical point of view, unlike the F/E muscles [such as flexor digitorum profundus (FDP) and flexor pollicis longus (FPL) muscles] distributed in the forearm, as shown in Fig. 2, the Ad/Ab muscles (such as adductor pollicis, palmar, and dorsal interosseous muscles) are basically distributed in the palm [37]. In Ad/Ab movement among digits, there exists coordination from the Ad/Ab relative to the neutral digit and the radial-ulnar deviation [38].

Coordination and independence between digits: The thumb is the most independent among the digits. The four fingers share an important flexor muscle, FDP muscle, while the thumb has a separate long flexor muscle (FPL muscle) in the forearm. In addition, the other multitendon muscles only connect the fingers, while the movements of the thumb joints are driven by exclusive muscles. The separate biomechanical structure is reflected in the poor movement-coordinated relationships between the thumb and the fingers.

TABLE I
STRUCTURE PARAMETERS OF DESIGNED HAND

Parameters	l_0	l_1	l_2	l_3
Values	50.8	52	35	35

Unit of measurement: *mm.*

Coupling and coordination within the fingers: Within each finger, correlations were low with the MCP joint and very high between the PIP and DIP joints. The flexion motions of the MCP, DIP, and PIP joints of each finger are primarily due to the separate actions of the FDP and FDS muscles. The compound tendinous attachments of extensor expansion connect the PIP and DIP joints and make the two joints rotate as a mechanism with one DOF when the FDP is active [39]. Except for some muscles such as the FDP and FDS sharing for the PIP and DIP joints, the lumbrical muscles are used exclusively for the MCP joints. It results in the relative independence of the MCP joint in the four fingers. These anatomical structures lead to the PIP and DIP joints of the four fingers more coordinated than the MCP joins.

It is worth noting that the musculoskeletal system is far more complicated than described above. What has been described above is the main aspect related to synergy. The biomechanical characteristics of hand synergy could serve as the basis for the transmission design of robot hands to extend the synergy-inspired design from anthropomorphic hands to general robot hands.

III. DESIGN

A. Hand Topological Structure

According to the biomechanical characteristics of human hand synergy, we design the transmission of a three-fingered hand. The designed hand structure, as shown in Fig. 3, is not anthropomorphic but symmetrical, like most three-fingered robot hands. The three digits are symmetrically distributed on the palm, and have the same structure and size. In details, the bases of the three digits are located on the three vertices of an equilateral triangle with side length l_0. Each digit has a joint for Ad/Ab and three joints for F/E. The lengths of three knuckles are l_1, l_2, and l_3, respectively. The parameter values of the structure are shown in Table I.

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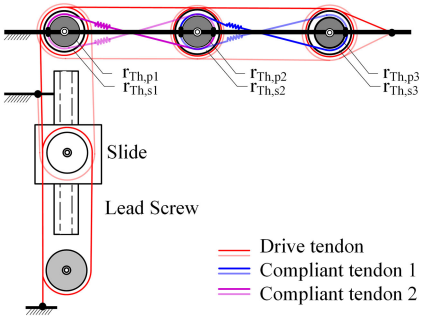


Fig. 4. Transmission for F/E of the thumb.

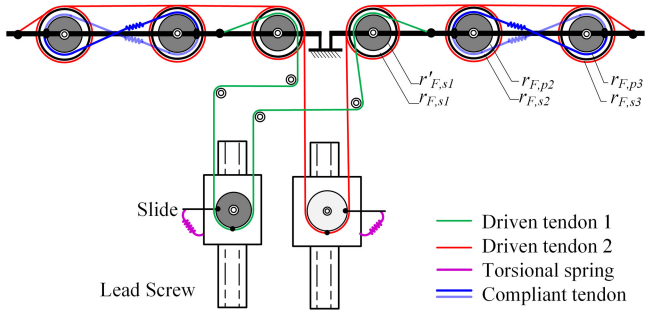


Fig. 5. Transmission for F/E of the two fingers. Note that the extension drive tendons, opposite to the corresponding flexion ones, are not appeared.

B. Transmission Mechanism

As mentioned in Section II, there are three biomechanical characteristics of hand synergy: The Ad/Ab movement is independent with F/E movement; The thumb is independent with the other fingers, while the corresponding joints of different fingers movements synchronously. The PIP and DIP joints of a finger more coordinated than the MCP joint. In our robot hand, these three characteristics are mapped into transmission mechanisms as three modules, namely the three Ad/Ab joints, the F/E joints of the thumb, and the F/E joints of the two fingers. These three modules are designed below.

The transmission for F/E joints of three digits is designed with a compliant underactuated mechanism. According to the transmission properties of motion and force in the compliant underactuated mechanism [40], differential mechanism with compliant couple (DM-CC) has low energy consumption and good energy efficiency. In the DM-CC, the active force and the passive force is provided by the serial transmissions and the parallel transmissions, respectively. The regulation of stiffness is nothing to do with the directions of motion and force. Hence, the transmission for F/E joints of three digits is designed with tendon-pulley DM-CC. The three F/E joints of the thumb are driven by an actuator through a tendon-driven differential transmission with two compliant couples, shown in Fig. 4. It has a pair of driven tendons inserted across three joints and two pair of passive elastic tendons compliantly coupling the three joints. This transmission mechanism was introduced in [23] for replacing the motion of the human digits. This mechanism has a stable motion trajectory before contacting with a grasped object. After contacting with the object, the finger is self-adapted by the contact constraint. In the enveloping grasping process, the joint trajectories tend to converge as the actuator displacement continues increase [41]. An optimization method of geometrical and elastic parameters was given in [41]. According to the characteristics of joint F/E motion within and between fingers elaborated in Section II, the F/E module of the two fingers includes six F/E joints driven by two actuators through two tendon-driven differential transmissions with four compliant couples, shown in Fig. 5.

According to the Ad/Ab movement characteristics of human hand in Section II, to carry out the coordination of the Ad/Ab relative to the neutral digit and radial-ulnar deviation, this module should achieve the counter movement of the two fingers

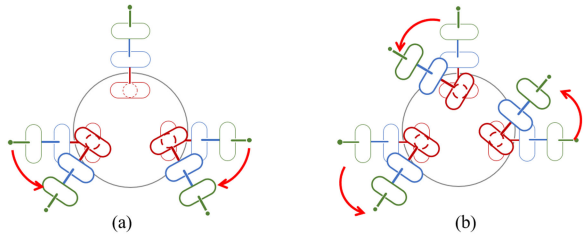


Fig. 6. (a) Reverse of the two fingers, and (b) homodromy of the three digits.

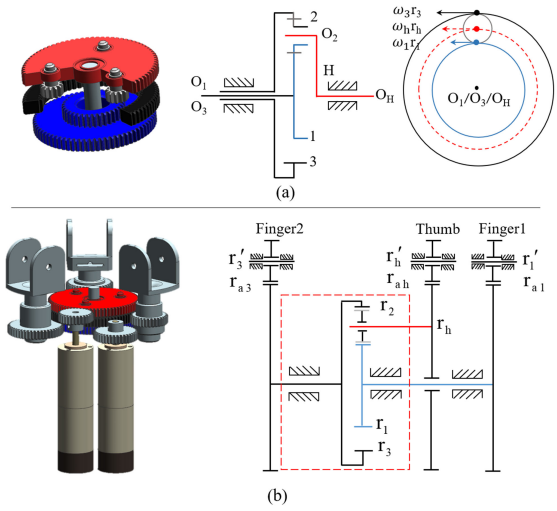


Fig. 7. (a) Principle of planetary gear system and its linear velocity relation. (b) Gear differential mechanism for two-DOFs Ad/Ab movements of three digits.

in Fig. 6(a) and the conjugate movement of the three digits in Fig. 6(b). Meanwhile, the two movements generate abundant finger postures via linear combination. The three Ad/Ab joints, $\mathbf{q}_A = [q_{A,T}\ q_{A,F1}\ q_{A,F2}]^T$, are driven by two actuators $\mathbf{a}_A = [a_{A1}\ a_{A2}]^T$ via a two-DOFs gear differential mechanism shown in Fig. 7. Thus, the two movement directions shown in Fig. 6 can be expressed with two vectors, namely $\mathbf{D}_1 = [1\ 1\ 1]^T$ and $\mathbf{D}_2 = [0\ 1\ -1]^T$ called *eigen directions* (EDs). And the normal vector of EDs is [-2, 1, 1]^T. Therefore, the constraint

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TABLE II
PARAMETERS OF GEAR DIFFERENTIAL MECHANISM

Parameters	r_1	r_1'	r_h	r_h'	r_3	r_3'	r_2	r_{a1}	r_{ah}	r_{a3}
Values	9	18	12	20	15	20	3	12	10	8

Unit of measurement: *mm*.

equation of the Ad/Ab movement can be obtained as

$$-2q_T+q_{F1}+q_{F2}=0 \tag{1}$$

which can be transformed into a planetary gear train in Fig. 7(a). The motion of the planetary gear train satisfies

$$2r_h\omega_h = r_1\omega_1 + r_3\omega_3 \tag{2}$$

where $\omega_h, \omega_1, \omega_3, r_h, r_1$, and r_3 are the angular velocities and the radius of connecting rod H, gear 1 and gear ring 3, respectively. Both sides of (2) are multiplied by time t, and one can get

$$2r_h\theta_h = r_1\theta_1 + r_3\theta_3. \tag{3}$$

To meet the planetary gear train with (1), a layer of external meshing gears is added to the planetary gear system as base gears of the three digits to drive the Ad/Ab motion as shown in Fig. 7(b). The radii of the base gears are denoted as r_1', r_h', and r_3', respectively. Then the mapping between $[\,q_T\ q_{F1}\ q_{F2}\,]^T$ and $[\,\theta_h\ \theta_1\ \theta_3\,]^T$ can be expressed as

$$\begin{cases} \theta_h = -q_T r_{ah}/r_h' \\ \theta_1 = -q_{F1} r_{a1}/r_1' \\ \theta_3 = -q_{F2} r_{a3}/r_3'. \end{cases} \tag{4}$$

Therefore, the constraint equation (3) can be expressed as

$$\frac{2r_{ah}r_h}{r_h'}q_T = \frac{r_{a1}r_1}{r_1'}q_{F1} + \frac{r_{a3}r_3}{r_3'}q_{F2}. \tag{5}$$

Comparing (1) and (5), the relation of gear radius needs to be proportional

$$\frac{r_{ah}r_h}{r_h'} : \frac{r_{a1}r_1}{r_1'} : \frac{r_{a3}r_3}{r_3'} = 1:1:1. \tag{6}$$

Meanwhile, the constraints of gear radius are restricted by

$$\begin{cases} 2r_h = r_1 + r_3 \\ 2r_2 = r_1 - r_3. \end{cases} \tag{7}$$

Considering the strength of the digit base and the hollow space of digit base need to through the rope, pipe thickness of the digit base should not be too small or it will affect the radius of the base gear connected to the digit base. Hence, the radius of the base gear cannot be less than 8 mm. Considering manufacturing constraints of r_2, the gear modulus is selected as 0.5. According to the above constraints, a group of feasible design parameters are obtained in Table II.

IV. GRASPING AND MANIPULATION ANALYSIS

In the designed hand, the new transmission mechanisms improve the ability of grasping and manipulation. This section presents a kinematic analysis of this hand for it.

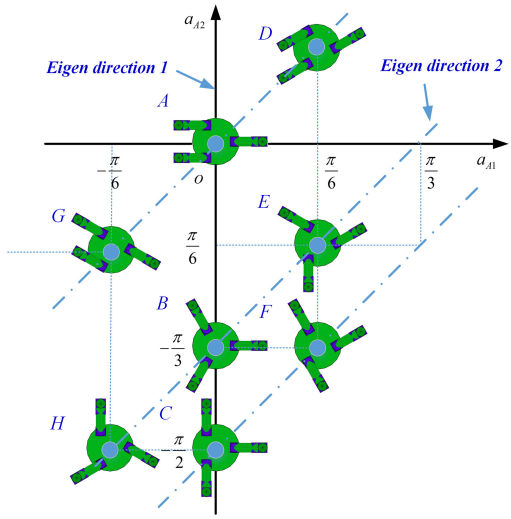


Fig. 8. Eigen directions and some postures of the Ad/Ab joins described in actuator space.

A. Hand Postures

As described in Section III, the transmission function of the Ad/Ab joints is designed as

$$\Delta\mathbf{q}_A = \mathbf{J}_A\Delta\mathbf{a}_A \tag{8}$$

where

$$\mathbf{J}_A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 2 & -1 \end{bmatrix}.$$

Meanwhile, the postures can be expressed with the linear combination of the two EDs

$$\Delta\mathbf{q}_A = \begin{bmatrix}\mathbf{D}_1\ \mathbf{D}_2\end{bmatrix}\begin{bmatrix}\Delta e_1 \\ \Delta e_2\end{bmatrix} \tag{9}$$

where e_1 and e_2 are the coordinates of the corresponding EDs, respectively. Let (9) be equal to (8), a map from the coordinates of the actuators to the ones of the EDs is gotten

$$\begin{cases} e_1 = a_{A2} \\ e_2 = a_{A1} + a_{A2}. \end{cases} \tag{10}$$

Hence, any reachable posture can be generated via the two EDs and their linear combination. Some typical postures are shown in Fig. 8.

The F/E joints of the thumb, $\mathbf{q}_{Th} = [\,q_{Th1}\ q_{Th2}\ q_{Th3}\,]^T$, are driven by an actuator a_{Th} through a tendon-driven differential transmission with two compliant couples shown in Fig. 4. The deformations of coupling springs are $\mathbf{T}_{Th,p} = [\,T_{Th,p1}\ T_{Th,p2}\,]^T$. The transmission functions are expressed as

$$\begin{cases} 2a_{Th} = r_{Th,s1}q_{Th1} + r_{Th,s2}q_{Th2} + r_{Th,s3}q_{Th3} \\ T_{Th,p1} = r_{Th,p1}q_{Th1} - r_{Th,p2}q_{Th2} \\ T_{Th,p2} = r_{Th,p2}q_{Th2} - r_{Th,p3}q_{Th3} \end{cases}$$

which is compactly rewritten as

$$\begin{bmatrix} 2a_{Th}\ \mathbf{T}_{Th}{}^T \end{bmatrix}^T = \mathbf{R}_{Th}\mathbf{q}_{Th} \tag{11}$$

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where

$$\mathbf{R}_{Th} = \begin{bmatrix} r_{Th,s1} & r_{Th,s2} & r_{Th,s3} \\ r_{Th,p1} & -r_{Th,p2} & 0 \\ 0 & r_{Th,p2} & -r_{Th,p3} \end{bmatrix}.$$

By solving the above linear equations, $\mathbf{q}_{Th}$ can be expressed as linear combination of a_{Th} and $\mathbf{T}_{Th,p}$

$$\mathbf{q}_{Th} = 2({\mathbf{R}_{Th}}^{-1})_{,1} a_{Th} + ({\mathbf{R}_{Th}}^{-1})_{,2:3} \mathbf{T}_{Th} \tag{12}$$

where $({\mathbf{R}_{Th}}^{-1})_{,1}$ is the first column of ${\mathbf{R}_{Th}}^{-1}$, and $({\mathbf{R}_{Th}}^{-1})_{,2:3}$ is the second and third columns.

The six F/E joints of the two fingers

$$\mathbf{q}_F = [\, q_{F1} \; q_{F2} \; q_{F3} \;\; q_{F4} \; q_{F5} \;\; q_{F6} \,]^T$$

driven by two actuators $\mathbf{a}_F = [\, a_{F1} \; a_{F2} \,]^T$ through two tendon-driven differential transmissions with four compliant couples shown in Fig. 5. The deformations of parallel springs

$$\mathbf{T}_{F,p} = [\, T_{F,p1} \; T_{F,p2} \; T_{F,p3} \; T_{F,p4} \,]^T$$

.

The transmission functions are expressed as

$$\begin{cases} 2a_{F_1} = r_{F,s1} q_{F1} + r_{F,s2} q_{F2} + r_{F,s3} q_{F3} \\ \quad + r_{F,s1} q_{F4} + r_{F,s2} q_{F5} + r_{F,s3} q_{F6} \\ 2a_{F_2} = {r'}_{F,s1} q_{F1} + {r'}_{F,s1} q_{F4} \\ T_{F,p1} = r_{F,s1} q_{F1} + r_{F,s2} q_{F2} + r_{F,s3} q_{F3} \\ \quad - r_{F,s1} q_{F4} - r_{F,s2} q_{F5} - r_{F,s3} q_{F6} \\ T_{F,p2} = {r'}_{F,s1} q_{F1} - {r'}_{F,s1} q_{F4} \\ T_{F,p3} = r_{F,p2} q_{F2} - r_{F,p3} q_{F3} \\ T_{F,p4} = r_{F,p2} q_{F5} - r_{F,p3} q_{F6} \end{cases}$$

which is compactly rewritten as

$$\left[{\mathbf{a}_F}^T \; {\mathbf{T}_{exF}}^T \; {\mathbf{T}_{inF}}^T \right]^T = \mathbf{R}_F \mathbf{q}_F \tag{13}$$

where $\mathbf{T}_{exF} = [\, T_{F,p1} \; T_{F,p2} \,]^{\mathrm{T}}$, $\mathbf{T}_{inF} = [\, T_{F,p3} \; T_{F,p4} \,]^{\mathrm{T}}$, and

$$\mathbf{R}_F = \begin{bmatrix} r_{F,s1} & r_{F,s2} & r_{F,s3} & r_{F,s1} & r_{F,s2} & r_{F,s3} \\ {r'}_{F,s1} & 0 & 0 & {r'}_{F,s1} & 0 & 0 \\ r_{F,s1} & r_{F,s2} & r_{F,s3} & -r_{F,s1} & -r_{F,s2} & -r_{F,s3} \\ {r'}_{F,s1} & 0 & 0 & -{r'}_{F,s1} & 0 & 0 \\ 0 & r_{F,p2} & -r_{F,p3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & r_{F,p2} & -r_{F,p3} \end{bmatrix}.$$

Then, $\mathbf{q}_F$ is a linear combination of $\mathbf{a}_F$, $\mathbf{T}_{exF}$, and $\mathbf{T}_{inF}$ as

$$\mathbf{q}_F = 2({\mathbf{R}_F}^{-1})_{,1:2} \mathbf{a}_F + ({\mathbf{R}_F}^{-1})_{,3:4} \mathbf{T}_{exF} + ({\mathbf{R}_F}^{-1})_{,5:6} \mathbf{T}_{inF}. \tag{14}$$

According to (12) and (14), the F/E joint space is spanned with active subspace driven by three actuators and passive subspace constrained with six groups of springs. During free motion before contacting with environment, the thumb moves along the direction of the vector $({\mathbf{R}_{Th}}^{-1})_{,1}$ and the fingers move in a plane spanned by the first two columns of ${\mathbf{R}_F}^{-1}$. In detail, any F/E motion of the two fingers is the combination of two independent motion, namely the F/E of MCP and the proportional motion of PIP and DIP as show in Fig. 9. After contacting with environment, the thumb and fingers can adapt to a grasped object via the elastic deformation of coupling springs along the subspace spanned by the columns of $({\mathbf{R}_{Th}}^{-1})_{,2:3}$, $({\mathbf{R}_F}^{-1})_{,3:4}$, and $({\mathbf{R}_F}^{-1})_{,5:6}$ in the F/E joint space.

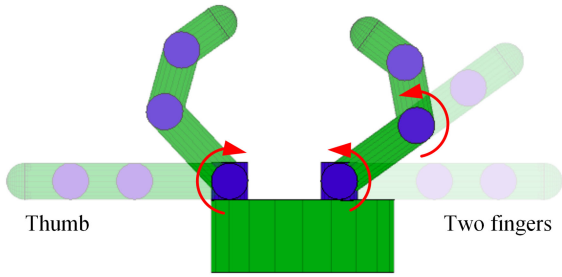


Fig. 9. Three independent motion of the F/E of the three digits.

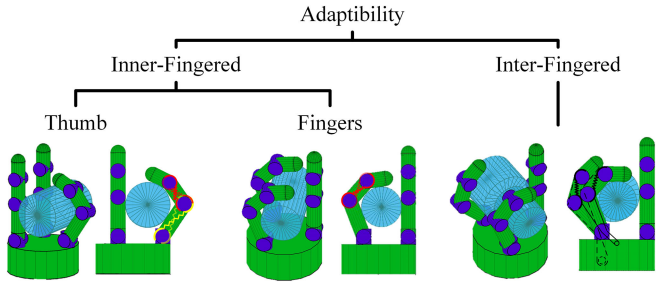


Fig. 10. Adaptive grasping taxonomy of the designed hand.

B. Dexterous and Adaptive Grasping

The grasp ability of this hand is analyzed from two aspects, namely dexterity and adaptability.

Grasp dexterity is mainly reflected in the ability of a hand to actively generate grasping postures. The grasp dexterity of the designed hand can be expressed with a five-dimensional (5-D) submanifold of joint space mapped from actuator space. As analyzed above, Ad/Ab postures can be achieved through the combination of the two EDs, shown in Fig. 8. In particular, some postures with different angles between the two fingers are generated via the ED1, while under the unchanged relative angles, the three digits synchronously rotate along the ED2. The F/E postures of the thumb are on the line along the direction of the vector $({\mathbf{R}_{Th}}^{-1})_{,1}$, and the F/E posture of the two fingers are in the plane spanned by the first two columns of ${\mathbf{R}_F}^{-1}$, as shown in Fig. 9. The combination of Ad/Ab and F/E motion can generate abundant grasping postures.

Adaptability means that the hand can adapt passively to the shape of the grasped object during grasping. The adaptability of the designed hand is reflected in the compliance of the DM-CC transmission mechanisms shown in Fig. 4 and (5). In detail, the adaptability includes inner-fingered adaptability as $\mathbf{T}_{Th}$ and $\mathbf{T}_{inF}$ expressed in (12) and (14), respectively, and inter-fingered adaptability as $\mathbf{T}_{exF}$ expressed in (14). Typical adaptive grasps are shown in Fig. 10. Inner-fingered adaptability is guaranteed by the spring couples between the three joints of the thumb and the ones between the PIP-DIP joints of the two fingers, while the differential transmissions between Finger1 and Finger2 gives the hand interfingered adaptability. The springs can adjust the finger joint motion with small deformation. The inner-fingered

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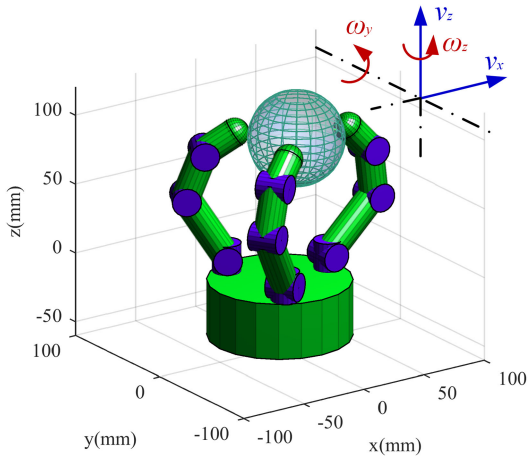


Fig. 11. Tripod manipulation with three fingertips.

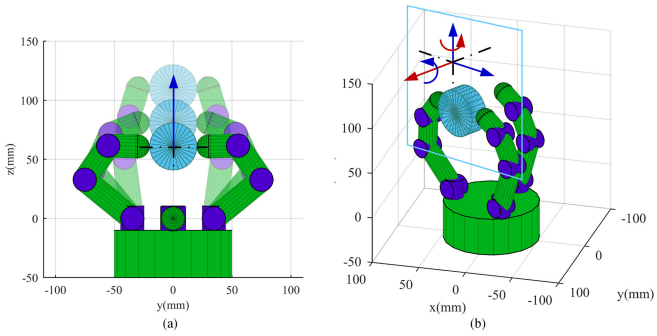


Fig. 12. Two-fingered manipulation (a) with Finger1 and Finger2, and (b) with the thumb and Finger1.

and inter-fingered adaptability allows the hand to adapt passively to objects with complex shapes in 3-D space. The dexterous and adaptive graspings are demonstrated in Section V.

C. In-Hand Manipulation

Usually, in-hand manipulation is carried out based on fingertip grasps. There are two manipulation types with the three-fingered hand, namely tripod manipulation and two-fingered manipulation.

A typical tripod grasp with three fingertips is shown Fig. 11, where the Ad/Ab is on Posture B in Fig. 8. Since the number of actuators is less than six, it is difficult for the hand to manipulation the object within hand in any pose. Because of the symmetrical arrangement of the two fingers and their transmission mechanism, the hand cannot manipulate the object to translate along the *Y*-axis and to rotate around *X*-axis. The hand can manipulate the object to realize the rotation around *Y*-axis and *Z*-axis and the translation along *X*-axis and *Z*-axis, as shown by the arrowed lines and arcs in Fig. 11. Besides, it can squeeze it while keeping the object still.

To carry out two-fingered manipulation, the hand can use two types of grasp postures as shown in Fig. 12. In Fig. 12(a), the Ad/Ab maintains Posture C in Fig. 8, and the F/E of the two fingers is driven by Actuators a_{F1} and a_{F2} via the transmission in Fig. 5. Hence, the two fingers can manipulate the grasped

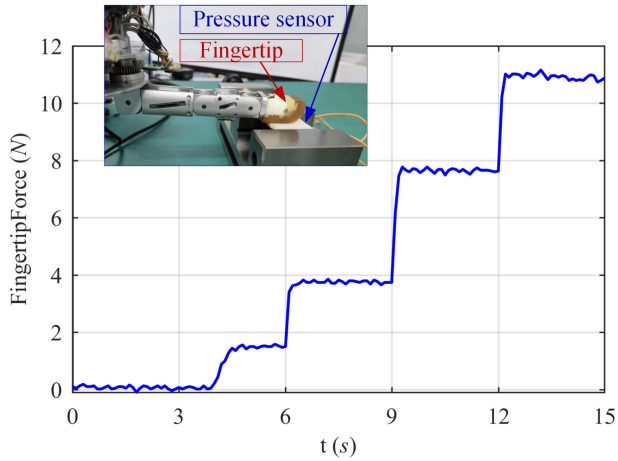


Fig. 13. Fingertip force response to the motor current.

object to move along the *Z*-axis while controlling the contact force. Compared with the conventional planar manipulation with one driver for each finger such as in [42], this cooperatively driven manipulation can achieve effective force control despite the sacrifice of operating space. It has potential application in alignment-insertion assembly task requiring precise force control. In Fig. 12(b), Posture G in Fig. 8, the grasped object is manipulated by the thumb and Finger 1. The manipulation system is driven by five actuators. The F/E of the thumb and Finger1 driven by three actuators a_{Th}, a_{F1}, and a_{F2} can realize the manipulation of three DOFs of the object in the F/E plane. The Ad/Ab of the two digits can be driven independently by two actuators a_{A1} and a_{A2} to operate the object moving perpendicular to the F/E plane and rotating about the *Z*-axis, as shown by the blue arrowed line and arc in Fig. 12(b). Since the two fingers are synchronously coupled, the hand cannot rotate the grasped object around the *X*-axis.

V. Experiments and Discussion

A. Synergy-Inspired Three-Fingered Hand Prototype

Based on the design and analysis in the above, a synergy-inspired three-fingered hand prototype has been developed, as shown in Fig. 1. This hand has 12 joints driven by five motors. Three dc motors (Maxon DCX19S, gear GPX19 6.6:1, and encoder ENX16) are used to drive the three ball screws (TBI SKF00401) as shown in Figs. 4 and 5, respectively, to achieve F/E of the three fingers. Two dc motors (Maxon DCX16S, gear GPX16 44:1, and encoder ENX16) are used to drive the gear differential mechanism in Fig. 7 to realize the Ad/Ab movements of the three digits. The hand spends 1.2 s from open to close.

The performance of the load capacity is evaluated by the fingertip force test. The fingertip force is defined as the positive fingertip pressure in the finger flat extension state [43]. Fig. 13 shows the experimental setup and results of the thumb fingertip force test. As the current increases stepwise from 0 to 1.6 A, the fingertip force also increases until 10.9 N. The maximal fingertip force is not less than 10 N. Meanwhile, in the designed hand phototype, the grasping forces vary with the change of

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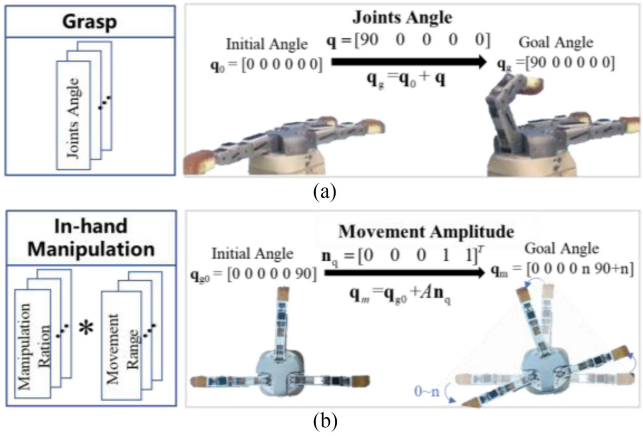


Fig. 14. Control commands for (a) grasp and (b) in-hand manipulation tasks.

the current of the motors. It means that the limitations of the grasping force are determined by the motor size rather than the hand construction.

In the experiments to carry out grasping and in-hand manipulation tests, we use only open-loop position control and current-limiting control on the five motors of the hand via proprioception sensors of encoders and current detectors. The control commands and communication flows during the initial and goal postures are shown with a block diagram in Fig. 14. During grasping, $\mathbf{q}_0 = [0\ 0\ 0\ 0\ 0]^T$ is the initial joint angle of thumb F/E, fingers MCP F/E, fingers PIP F/E, thumb Ad/Ab, and fingers Ad/Ab, respectively. By receiving a joint command, $\mathbf{q}$, the prototype can reach the desired posture, $\mathbf{q}_g = \mathbf{q}_0 + \mathbf{q}$, at a constant speed until the current of any a motor reaches the limited. During in-hand manipulation tasks, the motors are driven synchronously in a fixed proportion $\mathbf{n}_q$set empirically from a scheduled initial grasp posture $\mathbf{q}_{g0}$. The movement amplitude, A, is obtained by trial step by step until the contact constraints are broken like out of contact or grasping failure. Thus, the desired manipulation angle can be updated as $\mathbf{q}_m = \mathbf{q}_{g0} + A\mathbf{n}_q$.

B. Dexterity of Grasping

A quantitative analysis of grasp quality has been exposed in previous article [31] using the anthropomorphic hand assessment protocol [44]. This article aims to further explore the dexterity of the designed hand for common grasping tasks, therefore the Feix taxonomy [45] is used and the special postures in eigenspace directions are exposed. The Feix taxonomy includes 33 grasp types covering the grasps most common in human. There are some grasp types involving different numbers of fingers such as the *sphere 3-* and *4-finger, prismatic 2-, 3-,* and *4- finger, tripod, quadpod,* and *precision sphere* grasps. These are compressed to the *sphere 3-digit* grasp, the *prismatic 2-finger* grasp, and the *tripod* grasp in the test, respectively. Meanwhile, the *tripod variation* grasp involves five digits, so it is not included in the test. In conclusion, 27 grasp types are used for testing the designed hand. The experiment resulting postures are shown in Fig. 15. Out of 27 grasp types, the hand can perform 26 types. The complete grasping sequences are shown in the attachment material. Through the experiments, it is expected that the enhanced dexterity can be applied to various tasks in factory automation and daily life.

TABLE III
CONTROL COMMANDS OF MOTOR IN IN-HAND MANIPULATION TEST

Number of manipulation test in Fig. 16	Initial Joint Angle (q_{g0})	In-hand Manipulation Ratio(n_q)	Motion Amplitude (A)	Manipulation range (Mean ±error)
(a)	[45 90 60 30 30]	[1 0 -1 0 0]	0~23	64°±2°
(b)	[45 75 55 30 30]	[1 -2.5 0.5 0 0]	0~30	100±4mm
(c)	[0 90 30 0 90]	[0 -1 1 0 0]	0~60	51.5±0.5mm
(d)	[60 50 70 0 30]	[0 0 0 1 1]	-30~30	35°±2°

C. In-Hand Manipulation

The aim of the experiments is to verify the in-hand manipulation ability anticipated in Section IV-C. The experiments involved four typical in-hand manipulation tests, namely the translation and rotation in the F/E plane of the thumb and Finger 1 shown in Fig. 12(b), the translation along the Z-axis with Finger 1 and Finger 2, and the rotation around Z-axis with three digits. The manipulation objects are spheres whose radius is 25 mm for the first three tests, and 30 mm for the last one.

In an in-hand manipulation process, there are four stages, namely initial grasp postures implemented, control commands received, actuator displacement increased/decreased in fixed proportion synchronously, and maximum range of motion achieved. For each manipulation task, the above steps repeated three times under fixed movement ratio and amplitude. And the measured values of the three trials was averaged to calculate the repeatability errors. The repeatability errors were calculated by deviation between the extremum and means. The experimental test results are shown in Fig. 16, and the corresponding control commands are given in Table III. The complete manipulation sequences are shown in the attachment material. Specifically, the translation and rotation manipulations in the F/E plane of the thumb and Finger 1 are driven jointly by three flex-extension motors while the two Ad/Ab motors remaining stationary. The range of the object rotational displacement in Fig. 16(a),$\bar{\theta}_{\mathrm{tf}}$±error, is 64°±2°, as shown in Table III. The repetition error may be caused by the initial contact state, joint friction, and viscous deformation of elastic elements. The experiment results indicate that the hand prototype can perform a wide range of planar or spatial in-hand manipulation on a grasped object by simple synchronous proportional position control of motors.

D. Discussion

Table IV shows main parameters of the designed hand and some existing typical three-fingered hands. Compared with the other hands, our hand is moderate in weight, size, and the number of motors, with the largest number of joints. The most prominent feature of our hand is that the grouping and coordination design

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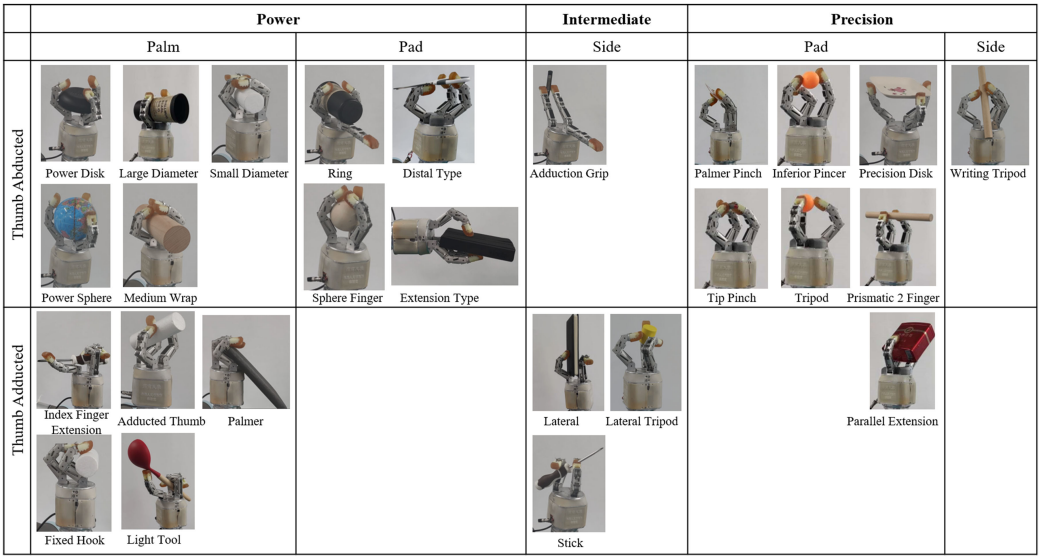


Fig. 15. Enacted grasps of the Feix taxonomy, using empirically determined actuation patterns.

TABLE IV
PARAMETERS OF TYPICAL THREE-FINGERED HANDS

		BarrettHand [7]	Robotiq 3-Finger Gripper [8]	iHYrobot Hand [46]	Schunk SDH [47]	Kinova Gripper KG-3 [48]	Our hand
Number of joints		8	10	8	8	6	12
Number of actuators		4	4	5	7	3	5
Arrangement of actuators		1-abd, 3-fe	1-abd, 3-fe	2-abd, 3-fe	1-abd, 6-fe	0-abd, 3-fe	2-abd, 3-fe
Motor Type		DC motors	DC motors	DC motors	DC motors	Linear actuators	DC motors
Type of coupling Mechanisms		Torque switch, worm drives, cable drive.	Gears and links	Tendon	Worm gears	Cable drive	Gears and cable drive
Size	Overall	187mm	155mm(Opening)	200mm	253mm	100mm(Opening)	232mm
	Finger	129mm		115mm	155mm		122mm
Weight		0.98kg	2.3 kg	1.35kg	1.95kg	0.727kg	1.22kg

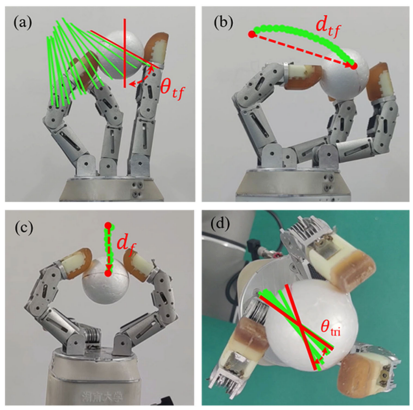


Fig. 16. Four typical in-hand manipulations and their range of motion. (a) Translation and (b) rotation in the F/E plane of the thumb and Finger 1, (c) translation along the *Z*-axis with two fingers, and (d) rotation around *Z*-axis with three digits.

of transmission mechanism inspired by the crucial biomechanical characteristics of human hand synergy which giving the hand greater dexterity and adaptability.

The grasping dexterity of the designed hand can be clarified from the perspective of the combination of Ad/Ab and F/E as shown in Fig. 17. As with most of the existing three-fingered hands such as the ones in Table IV, the designed hand can achieve two-finger symmetrical rotation, i.e., ED1 in Fig. 8. Particularly, postures A, B, and C in Fig. 8 are the most common Ad/Ab postures for grasping, as shown in the top three rows of Fig. 17. Distinctively, some unique grasp posture can be generated for our hand. For example, ED 2 in Fig. 8 leads to some new grasping modes such as fingertip grasping a ball with the thumb and a finger in Grasp D3, side pinching a key in Grasp F4 in the bottom three rows of Fig. 17 and envelop grasp for small object using fingertip and part of finger surface/side in the row of E. The coordinated F/E movement between Finger 1 and Finger 2 does not only improve the dexterity such as Grasp F1 and F2, but also improves the adaptability such as Grasp A2 in Fig. 17. Through the coordination of Ad/Ab and F/E, the hand can generate a lot of practical grasping postures, which enriches greatly the diversity of grasping modes. The compliant coupling structure and synergy transmission in the hand do not only reduce the control complexity, but also increase the dexterity of the robotic hand.

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Fig. 17. Grasps from the perspective of the combination of Ad/Ab and F/E. Each row corresponds to an Ad/Ab posture.

The failed one of the 27 grasp types in Section V-B is the *ventral* grasp. This is because the thumb cannot swing independently to the same side of the other two fingers. It should be noted that due to the loss of the independence of the thumb's Ad/Ab movement, there are several grasps of the designed hand different from human hands, such as the *lateral, fixed hook, and palmer* grasps. For some grasps, the designed hand has a variety of implementation methods. For example, the *tip pinch* grasp can be achieved with the thumb and a finger shown in Grasp D4 of Fig. 17, or with the two fingers shown in Grasp C4 of Fig. 17. Attributing to the coordinated F/E transmission between Finger 1 and Finger 2, the *distal type* grasp can be used to operate scissors to cut things.

The hand is expected to be used in space station maintenance, human–machine collaboration and other scenarios with high requirements for dexterous grasping and manipulation. It is worth noting that, in the experimental results in Section V-C, the in-hand manipulation accuracy is not high. It is mainly due to open loop control in our verification experiment on the design method. A closed-loop control based on touch sensors and visual sensors could significantly improve the control accuracy of the system. Meanwhile, the manipulation amplitude given in the experimental results is only a feasible one, other than the maximum manipulation amplitude of the hand. To improve the manipulation performance, the appropriate control strategy is worthy of further study, involving tactile visual fusion and reinforcement learning.

VI. Conclusion

This article presents a design methodology of multifingered robotic hands to replicate the dexterous grasping and in-hand manipulation functions of the human hand by reconstructing mechanically the crucial biomechanical characteristics of human hand synergy. To formulate such a methodology, the human hand synergy is reexamined from the perspective of biomechanics. Through the comparative analysis of human hand movement characteristics from hand biomechanics and joint data statistics, the human hand synergy is summarized as the Ad/Ab movement of the digits is independent with F/E movement; the F/E of thumb is independent with the other fingers, while the corresponding joints of different fingers movements synchronously; the PIP and DIP joints of a finger more coordinated than the MCP joint. These basic characteristics of human hand synergy make it not only possible but also very convenient to implement mechanically in robot hands with various topological structures, not just in anthropomorphic structures.

Based on the biomechanical characteristics of hand synergy, a compliant, underactuated, and dexterous three-fingered hand has been developed. Its kinematic model is presented, and the typical grasping and manipulation functions are analyzed. The hand has two remarkable characteristics, namely the coordination of the two Ad/Ab movements and the compliant differential transmission between the fingers. From the results of experimental tests, we see that the designed hand can achieve dexterous grasping and in-hand manipulation function with a simple control scheme. By comparing with the existing commercial three-fingered hands, it is not difficult to find that the designed hand is equivalent to other hands in size, weight, and number of motors. Due to the bionic design of the transmission system, the dexterity of our hand is unmatched by other hands, as shown in the results of experimental tests. Simulation analysis and experimental tests verify the effectiveness of the design method.

The potential application of the design principle is the universal robot end-effector for complex environments such domestic nursing, aerospace, and nuclear industry. To improve the robust and economic quality of the synergy-inspired hand, flexible joints and soft structure can be considered. In view of the results of empirical open-loop position control in the experiments, it is necessary to develop closed-loop control algorithms integrated visual and tactile sensors to enhance its dexterity and accuracy in future article.

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